DETAILED DESCRIPTION OF THE INVENTION

Example Embodiment Walkthrough

In one illustrative embodiment, a handheld penile anatomical geometry profiling device comprises a semi-rigid housing incorporating four time-of-flight (ToF) ranging sensors positioned at angular offsets within a C-shaped frame and two optical motion sensing components positioned along a longitudinal contact pathway. The ToF sensors are oriented toward the anatomical surface and configured to capture radial distance measurements relative to a defined reference plane within the housing structure.

Prior to use, the device performs a startup calibration routine verifying baseline distance offsets and motion sensor responsiveness. Upon placement at a defined anatomical starting position, the system confirms tissue presence through a pressure-based compliance sensor and optional photoplethysmography (PPG) detection of vascular pulsatility. Once validation thresholds are satisfied, the user initiates a longitudinal scanning motion along the anatomical axis.

During scanning, radial distance samples from the four ToF sensors are captured at high frequency and temporally synchronized with displacement data obtained from the optical motion sensors. Inertial measurement data is concurrently collected to compensate for tilt or unintended angular deviation. The synchronized dataset is encrypted in memory and transmitted to an onboard reconstruction engine.

The reconstruction engine interpolates intermediate surface points between sampled positions, constructs discrete cross-sectional profiles at defined longitudinal intervals, and generates a continuous three-dimensional surface model. Curvature vectors are calculated by analyzing centroid shifts between cross-sections, and volumetric estimation is derived through numerical integration of cross-sectional areas.

Derived structural metrics including total longitudinal length, radial distribution curves, taper gradients, symmetry indices, and proportionality ratios are computed and encoded into a standardized digital geometry profile. The encoded profile is stored in encrypted form and may optionally be transmitted to a companion application for comparative modeling, device calibration, or integration within a broader biometric ecosystem.

This example embodiment is provided for illustrative purposes and does not limit the disclosure to the specific sensor count, housing configuration, sampling rate, reconstruction algorithm, encryption method, or deployment architecture described. Variations in hardware arrangement, data processing methodology, and integration model remain within the scope of generating a user-specific penile anatomical geometry profile using synchronized time-of-flight distance and motion measurements.